Each Lightning Bug was equipped with high-resolution film cameras, radar sensors, and telemetry systems, capable of autonomous, pre-programed flight at altitudes above 60,000 feet. After mission completion, drones were recovered by parachute and retrieved mid-air by helicopters or aircraft to preserve the valuable reconnaissance film.

The data collected by these UAVs was instrumental in battle damage assessment, strategic target identification, and air defense mapping. The Lightning Bug's success showcased the feasibility of pilotless reconnaissance, significantly reducing human risk. This program validated UAV technology as a cost-effective, survivable alternative to manned spy aircraft, influencing future drone programs such as the RQ-2 Pioneer and Global Hawk.

Impact:

- Introduced the concept of reusable UAVs for long-range intelligence.
- Demonstrated high-altitude, autonomous navigation with precision camera payloads.
- Established drones as critical tools for intelligence, surveillance, and reconnaissance (ISR).

3.7.2 Case study 2: IAI Scout – real-time intelligence in the Lebanon War (1982)

The IAI Scout, developed by Israel Aerospace Industries, was the first UAV to integrate real-time video transmission and electronic warfare (EW) payloads into a combat-proven platform. During the 1982 Lebanon War, the Scout provided live imagery of enemy movements and coordinated strikes with manned aircraft and artillery units, revolutionizing the use of drones in tactical warfare.